

Mathematical Notation Preview

Một số công thức trong FastSLAM

Xác suất có điều kiện:

$$p(x_t \mid x_{t-1}, u_t)$$

Phân phối Gaussian:

$$\mathbf{x} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$$

Innovation:

$$\mathbf{y}_t = \mathbf{z}_t - h(\boldsymbol{\mu}_t)$$

Innovation covariance:

$$\mathbf{S}_t = \mathbf{H}_t \boldsymbol{\Sigma}_t \mathbf{H}_t^\top + \mathbf{R}_t$$

Kalman gain:

$$\mathbf{K}_t = \boldsymbol{\Sigma}_t \mathbf{H}_t^\top \mathbf{S}_t^{-1}$$

EKF update:

$$\boldsymbol{\mu}_t^+ = \boldsymbol{\mu}_t^- + \mathbf{K}_t \mathbf{y}_t$$

Covariance update:

$$\boldsymbol{\Sigma}_t^+ = (\mathbf{I} - \mathbf{K}_t \mathbf{H}_t) \boldsymbol{\Sigma}_t^-$$

Mahalanobis distance:

$$d^2 = \mathbf{y}^\top \mathbf{S}^{-1} \mathbf{y}$$

Pose 2D:

$$\mathbf{x}_t = \begin{bmatrix} x_t \\ y_t \\ \theta_t \end{bmatrix} \in SE(2)$$